

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS2

SUBJECT: Agricultural Science

TOPIC: Animal Health & Diseases

Introduction

- Animal health is the state of well-being of farm animals, free from disease. Disease reduces production, costs money and can spread to humans (zoonoses) — so prevention is the farmer's first duty.

Classes of diseases

- **Viral** — Newcastle disease (fowl plague), Gumboro, foot-and-mouth disease, African swine fever.
- **Bacterial** — fowl cholera, tuberculosis, anthrax, mastitis.
- **Protozoan/parasitic** — coccidiosis, trypanosomiasis (sleeping sickness), worm infestations (roundworms, tapeworms), tick-borne diseases.
- **Nutritional and metabolic** — deficiency diseases, bloat.

Signs of ill health

- Loss of appetite, dullness, drooping wings, diarrhoea, discharges, fever, loss of weight, difficulty in breathing, and reduced production (eggs, milk). Isolate sick animals at once.

Prevention and control

- Vaccination (Newcastle, Gumboro, anthrax); routine deworming and dipping/spraying against ticks; good hygiene (clean pens, clean water, dry litter); balanced feeding; quarantine of new animals; proper disposal of dead animals; and prompt veterinary attention. The testse-fly control programme reduces trypanosomiasis.

CLASS EVALUATION

1. Define animal health.
2. Name two viral and two bacterial diseases of farm animals.
3. State three signs of a sick animal.
4. Mention four ways of preventing animal diseases.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS2

SUBJECT: Agricultural Science

TOPIC: Farm Records & Accounting

Introduction

- Farm records are written accounts of the farm's activities; farm accounting is the recording and analysis of these figures in money terms. Records show whether the farm is making a profit and guide decisions.

Types of farm records

- **Inventory record** — the farm's assets (tools, livestock, machinery) and their values.
- **Production record** — crops harvested, eggs, milk and offspring produced.
- **Labour record** — days of work, wages paid.
- **Sales record** — produce sold, buyers, prices and dates.
- **Purchase record** — inputs bought (seed, feed, fertiliser, drugs).
- **Cash book / financial record** — all money in and out; **farm diary** — daily activities, weather and observations.

Farm accounting terms

- **Fixed costs** — rents, interest, depreciation (they do not change with output); **variable costs** — seed, feed, fertiliser, labour (they change with output); **total cost** = fixed + variable; **revenue** = sales; **profit** = revenue – total cost; **break-even point** — where revenue just covers cost.

Uses of records

- Planning, budgeting, credit applications, tax assessment, detecting losses and fraud, and comparing enterprises.

CLASS EVALUATION

1. Define a farm record.
2. List four farm records.
3. Distinguish between fixed and variable costs.
4. How is profit calculated and what is the break-even point?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS2

SUBJECT: Agricultural Science

TOPIC: Agricultural Marketing

Introduction

- Agricultural marketing is the process of moving farm produce from the producer to the consumer — assembling, grading, processing, packaging, transporting and selling.

Marketing functions

- **Assembling** — collecting produce from many small farms; **grading and standardising** — sorting by quality and size (higher grades fetch higher prices); **processing and packaging** — increasing shelf life and value; **storage** — holding produce for better prices; **transportation**; **financing** the trade; **risk-bearing** (price changes, spoilage); and **price discovery** (supply and demand meeting in the market).

Marketing channels

- Farmer → local assembler → wholesaler → retailer → consumer. The chain of middlemen takes a large share of the final price — often leaving farmers under half of what the consumer pays — hence the call for direct marketing and cooperatives.

Problems and solutions

- Problems: poor roads and storage, price instability, seasonality, middlemen exploitation, adulteration, lack of market information, pests and poor packaging. Solutions: market cooperatives, warehouses and silos, good rural roads, price stabilisation schemes, extension advice on grading, and the use of ICT for market information.

CLASS EVALUATION

1. Define agricultural marketing.
2. State five marketing functions.
3. Why do farmers get a small share of the consumer's price?
4. Mention three problems of agricultural marketing and one solution each.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS2

SUBJECT: Agricultural Science

TOPIC: Agricultural Extension

Introduction

- Agricultural extension is the service that educates farmers — the transfer of new ideas, improved practices and research findings from the laboratory to the farm. In Nigeria, the extension arm of the Agricultural Development Programmes (ADPs) with the World Bank and the National Agricultural Extension and Research Liaison Services (NAERLS) lead the work.

Functions of extension

- Teaching improved farming methods (use of fertiliser, improved seed, pest control); introducing new crops and animals; advising on marketing, credit and record keeping; training farmers through demonstrations, field days and farmers' groups; and collecting feedback on farmers' problems for research.

Methods of teaching

- Individual (farm visits, office calls); group (demonstrations, field days, farmers' training centres); and mass methods (radio, television, posters, the Farm Radio programmes, newsletters and mobile phone messages).

Problems of extension work

- Too few extension agents for many farmers; poor funding and transport; illiteracy and conservatism among farmers; communication and language gaps; and the gap between research and the farmer's reality.

CLASS EVALUATION

1. Define agricultural extension.
2. State four functions of the extension service.
3. Differentiate group from mass teaching methods.
4. Mention three problems of extension work in Nigeria.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS2

SUBJECT: Agricultural Science

TOPIC: Poultry & Livestock Production II (Ranching & Feed Formulation)

Introduction

- This week takes livestock production forward: ranching and the formulation of balanced feed — the two areas that determine the productivity of the Nigerian livestock industry.

Ranching and livestock systems

- **Ranching** — grazing cattle on fenced, managed natural or planted pasture, with controlled breeding, water and veterinary care (plateau and northern ranches); **nomadic herding** — moving with the herd in search of pasture and water (the traditional Fulani system, with the problems of overgrazing and conflict); **zero grazing** — cut-and-carry feeding in pens; **intensive** — fully housed (dairy, poultry, pigs) with complete feed.

Feed formulation

- A balanced ration supplies energy (maize, guinea corn), protein (soya, groundnut cake, fish meal), minerals (bone meal, limestone), vitamins and water in the right proportions. Formulation by the Pearson square: mix maize (10% protein) and soya meal (44%) to give a 20% ration — the square gives the ratio. Feeding is adjusted to the animal's age, type and purpose (growth, fattening, lactation, egg production).

CLASS EVALUATION

1. Define ranching.
2. State two problems of nomadic herding.
3. What are the components of a balanced feed?
4. Use the Pearson square to find the ratio of maize (9% CP) and groundnut cake (45% CP) for a 20% CP ration.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS2

SUBJECT: Agricultural Science

TOPIC: Fisheries & Bee Farming (Apiculture)

Introduction

- Fisheries and apiculture are the intensive forms of animal production we have not yet visited: fish farming (aquaculture) and bee keeping. Both are growing enterprises in Nigeria — fish for protein, bees for honey and wax.

Fish farming

- Fish are a cheap source of animal protein. Culture systems: **pond culture** (earthen, concrete and flow-through ponds), **cage and pen culture** (in rivers and dams), **raceways** (running water) and **aquaponics** (fish and vegetables together). Species: tilapia (the hardiest), catfish (Clarias — the most profitable), carp, heterotis and the mudfish.

The fish farm operations

- Site selection (a reliable water source, good soil); pond construction and liming; stocking (30–100 fish per square metre for catfish in tanks, fewer in ponds); feeding (high-protein compounded feed — 2–5% of body weight); water quality (dissolved oxygen, pH 6.5–8.5); sorting and grading; harvesting (partial and total); and processing: gutting, salting, smoking, drying and freezing.

Beekeeping (apiculture)

- The honeybee colony: the queen, workers and drones; the hive (the modern movable-frame hive), the bee space, frames and foundation. Siting: near flowers, away from noise and chemicals; the apiary routine: inspection, feeding (sugar syrup in the dearth), swarm control, and harvesting honey by the centrifugal extractor. Products: honey (food, medicine), beeswax, propolis and royal jelly; the bees pollinate crops — increasing yields.

CLASS EVALUATION

1. Define aquaculture.
2. Name four fish culture systems.
3. State four factors that determine good fish pond water.
4. Describe the members of a bee colony.
5. Why are bees important to crop production?

